

When a Decision Does Not Resolve

Non-resolution and persistent indeterminacy in AI judgement
systems

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Executive Summary

AI systems are often evaluated based on whether their outputs appear reasonable, consistent, or well explained. More recent approaches extend this by examining how those outputs behave when the same task is repeated. Where variation appears across repeated runs, this is typically interpreted as instability in judgement.

This paper identifies a different pattern.

In a subset of evaluative tasks, the system does not vary in its judgement under repeated execution, but nor does it resolve that judgement into a definitive classification. Instead, it consistently returns an indeterminate outcome.

Using repeated evaluations of job advertisements for potential age-related bias, the analysis shows that early-career roles containing signals such as “junior”, “graduate”, or “trainee” are classified as “Unclear” in every case. This result remains unchanged across repeated runs, with no movement into adjacent categories and no evidence of threshold crossing.

Confidence follows the same pattern. Values remain tightly clustered at moderate levels, with little variation and no increase in certainty over time.

This behaviour differs from the instability observed elsewhere in the dataset. In other cases, particularly at decision boundaries, judgements vary across runs, moving between adjacent categories such as “Potentially Biased” and “Unclear”. In contrast, the early-career subset shows no variation, but also no resolution. This distinction is important.

A stable output does not necessarily represent a resolved decision. In these cases, the system detects relevant signals but does not assign sufficient weight to reach a classification. The result is a consistent but indeterminate outcome.

This paper introduces non-resolution as a distinct behavioural property of AI judgement systems. It shows that boundary cases do not produce a single type of behaviour. Some fluctuate under repeated evaluation, while others remain fixed in an unresolved state.

For evaluation and audit, this has practical implications. Approaches that focus on variation alone may overlook cases where the system consistently fails to commit. Stability, when observed in isolation, may reflect persistent indeterminacy rather than reliable decision-making.

Reliability therefore depends not only on whether a judgement remains consistent, but on whether it resolves into a usable form.

Abstract

Most approaches to evaluating AI systems focus on the quality of individual outputs or, more recently, on how those outputs vary when the same task is repeated. While variation under repeated evaluation reveals important properties of system behaviour, it does not fully describe how judgements are formed in cases where signals are weak or ambiguous.

This paper examines a different behavioural pattern: non-resolution. In a subset of evaluative tasks, AI systems do not move between classifications under repeated execution, but instead

remain fixed in an indeterminate state. Using repeated evaluations of job advertisements for potential age-related bias, the analysis shows that certain inputs consistently produce “Unclear” outcomes across multiple runs, with stable confidence and no evidence of threshold crossing.

This behaviour is not instability in the usual sense. The system does not vary in its judgement, but nor does it resolve the judgement into a definitive classification. The findings suggest that non-resolution is a distinct behavioural property, reflecting the system’s sensitivity to weak or ambiguous signals and its inability to commit when decision thresholds are not met.

These results extend the behavioural evaluation approach by identifying persistent indeterminacy as a separate mode of system behaviour. Reliability in AI judgement systems must therefore account not only for whether decisions remain stable, but whether they resolve at all.

Keywords

- AI evaluation
- LLM evaluation
- model reliability
- judgement stability
- behavioural audit
- decision boundaries

1. Introduction

Evaluation of AI systems is often based on the inspection of individual outputs. A response that appears reasonable or well explained is frequently taken as evidence that the system is behaving reliably.

This assumption becomes less certain when the same task is observed repeatedly. Earlier work has shown that AI judgements do not always remain the same under repeated evaluation. Where interpretation is required, variation emerges, particularly at decision boundaries.

At those boundaries, however, behaviour does not follow a single pattern. In many cases, judgements move between adjacent categories across runs, reflecting sensitivity to marginal differences in how signals are interpreted.

In other cases, the behaviour is different. The judgement does not move, but nor does it resolve into a definitive classification. Instead, the system consistently returns an indeterminate outcome, maintaining the same position despite repeated evaluation under identical conditions.

This paper examines that pattern. Building on earlier work on behavioural evaluation and judgement stability, the analysis focuses on non-resolution as a distinct behavioural property. Rather than asking whether a judgement remains the same, the question becomes whether the system is able to resolve the judgement at all.

Earlier work identified variation in judgement at decision boundaries, where classifications move between adjacent categories under identical conditions. The present analysis examines a different pattern, where judgements remain stable but indeterminate.

Taken together, these findings indicate that boundary cases do not produce a single type of behaviour. Some vary under repeated evaluation, while others remain stable without resolving. Both patterns are relevant to understanding how AI systems behave under ambiguity.

2. Research Context

This paper forms part of the *Agents at Work* research series, which examines how AI systems interpret age-coded language in recruitment text. The specific focus on early-career roles emerged from discussion, which prompted closer examination of this subset within the dataset.

Earlier work introduced the Behavioural Evaluation Framework, which proposes assessing AI systems through repeated execution, internal signals, and comparison across independent systems. This framework reframed evaluation as the observation of behavioural patterns rather than the inspection of individual outputs.

Subsequent analysis examined judgement stability under repeated execution. While most classifications remained consistent, a meaningful proportion varied under identical conditions, with variation concentrated at decision boundaries.

The present paper extends this line of enquiry.

Where earlier work focused on variation in judgement, this analysis examines cases where variation does not occur, but resolution is also absent. The focus is therefore not on instability, but on persistent indeterminacy within the classification process.

3. Method Overview

The analysis is based on repeated evaluations of job advertisements for potential age-related bias.

Each advertisement was assessed multiple times using the same model, prompt, and input conditions. The classification task required assignment to one of four categories:

- Likely Biased
- Potentially Biased
- Unclear
- Unbiased

A subset of advertisements containing explicit early-career signals, such as “junior”, “graduate”, “trainee”, or “entry level”, was isolated for further analysis. This subset comprised 52 adverts within a broader sample of 150 UK job advertisements.

Each advert in the subset was evaluated across three repeated runs under identical conditions. The objective was to observe whether judgements remained stable, varied across runs, or resolved into more definitive classifications.

The focus of the analysis is not classification accuracy, but the behaviour of the system under repeated execution.

4. Findings

4.1 Persistent “Unclear” Classification

Across all 52 early-career adverts, the system produced the same classification in every run.

Each advert was classified as “Unclear”, and no case moved into “Potentially Biased”, “Likely Biased”, or “Unbiased”. This pattern held across all repeated evaluations, with no evidence of variation or threshold crossing and no movement toward more definitive categories over repeated runs.

4.2 Absence of Variation

Unlike other subsets of the dataset, these adverts did not exhibit instability. There was no movement between categories and no evidence of drift or convergence across runs.

Judgement remained fixed across repeated evaluation.

4.3 Confidence Behaviour

Confidence values remained tightly clustered across runs, typically centred around moderate levels. There was minimal variation in confidence and no instances of high-confidence classification.

This indicates that the system consistently expresses moderate certainty in an indeterminate outcome.

4.4 Contrast with Other Outputs

In the remaining adverts within the dataset, a different pattern was observed.

A proportion of cases exhibited variation across runs, particularly between “Potentially Biased” and “Unclear”. Some inputs produced more decisive classifications, and confidence values showed greater dispersion.

This contrast highlights a distinct behavioural difference. Some inputs produce instability at the boundary, while others produce persistent non-resolution.

5. Interpretation

The observed pattern does not indicate that the system is failing to repeat its judgement. Instead, it indicates that the system is unable to resolve the judgement.

Where signals are weak, mixed, or open to interpretation, the system detects relevant cues but does not assign sufficient weight to cross the threshold required for classification. As a result, the judgement remains in an indeterminate state.

This behaviour differs from instability.

In unstable cases, the system moves between adjacent categories across runs, reflecting sensitivity to marginal differences in internal processing. In non-resolving cases, the system does not move at all. It consistently returns the same indeterminate outcome.

This suggests that decision boundaries do not produce a single behavioural response. Instead, they give rise to at least two distinct patterns:

- variation across runs
- persistent non-resolution

The presence of non-resolution indicates that stability alone is not sufficient to characterise system behaviour. A judgement may remain stable while still failing to resolve into a meaningful classification.

6. Implications for AI Evaluation

These findings extend the interpretation of behavioural stability in AI systems.

First, stability should not be interpreted as evidence that a system has reached a definitive judgement. A stable output may represent a persistent lack of resolution rather than a reliable decision.

Second, evaluation approaches that focus on variation alone may overlook cases where the system consistently fails to commit. Non-resolution is not visible as instability, but it has practical consequences for how outputs are interpreted and used.

Third, the presence of persistent “Unclear” outcomes suggests that AI systems may handle weak or ambiguous signals by maintaining an intermediate position rather than expressing uncertainty explicitly or resolving the judgement.

In decision-making contexts, this behaviour may affect how borderline cases are treated. A system that consistently returns an indeterminate outcome may appear stable, but does not provide a basis for action.

Reliability therefore depends not only on whether a judgement remains consistent, but on whether it resolves into a usable form.

7. Limitations

This analysis focuses on a specific subset of recruitment text containing early-career signals. While the observed pattern is clear within this subset, further work is required to determine how widely non-resolution occurs across other domains and tasks.

The evaluation is conducted under controlled conditions using a fixed prompt and model configuration. Real-world use may introduce additional variability through changes in input structure or system settings.

The analysis does not assess classification accuracy. The purpose is to examine behavioural properties rather than determine whether individual outputs are correct.

8. Conclusion

When the same evaluative task is repeated, AI judgements do not always vary. In some cases, they do not resolve.

This paper identifies non-resolution as a distinct behavioural property of AI judgement systems. Rather than moving between classifications, the system remains fixed in an indeterminate state, consistently returning the same outcome across repeated evaluation.

This behaviour is not captured by approaches that focus on single outputs or on variation alone. A judgement may appear stable while failing to resolve into a definitive classification.

Understanding this distinction is essential for evaluating how AI systems behave in practice. Stability does not guarantee resolution. A judgement may remain consistent without resolving into a usable decision. Reliability depends not only on whether a judgement holds, but on whether it resolves at all.

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